

Muslim A. Alaran

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Education

- PhD in Robot Learning**, Montanuniversität Leoben, Leoben, Austria Aug 2026 – Expected 2030
- **Advisor:** Prof. Elmar Rückert.
- MS in Robotics**, Nazarbayev University, Astana, Kazakhstan Aug 2022 – Jun 2024
- GPA: 3.7/4.0 | Dean's Honor List, Spring, Fall 2023.
 - **Thesis:** Enhanced Object Recognition in Hemispherical Images using Data Augmentation and Synthetic Image Integration.
 - **Advisor:** Prof. Huseyin Atakan Varol.
- BEng in Electrical and Electronics Engineering**, University of Ilorin, Ilorin, Nigeria Oct 2015 – Aug 2021
- Grade: First Class Honors (Top 1% of graduating cohort).
 - **Thesis:** Attendance Monitoring System using RFID and Fingerprint.
 - **Advisor:** Dr. I.O.A Omeiza.

Research Experience

- Doctoral Researcher**, Chair of Cyber-Physical Systems, Montanuniversität Leoben, Austria Aug 2026 – Present
- Researching skill learning for humanoid robots (Unitree G1, R1) targeting industrial production and logistics tasks.
- Junior Researcher**, Artificial Intelligence for Robotics Lab, FER, University of Zagreb, Croatia Mar 2025 – Jul 2026
- Developed and deployed imitation and reinforcement learning policies for manipulation on Kinova Gen3 arms.
 - Fine-tuned Octo, a generalist robot policy, for goal-image-conditioned bottle pick-and-lift from ~110 demonstrations, using a hybrid action head.
 - Built the full real-robot pipeline (demonstration collection, hand-eye calibration, TFDS/RLDS conversion, offline evaluation, live deployment) around a latency-aware receding-horizon controller; released open-source (v1.0.0).
 - Owned sim-to-real transfer for a decomposed-reward RL architecture with spatially aware value fusion for marine manipulation.
- Research Assistant**, Robot Control and Learning Lab, Nazarbayev University, Kazakhstan Aug 2024 – Feb 2025
- Built a safety-critical reinforcement learning framework for human-robot collaboration, enforcing Speed and Separation Monitoring constraints in PPO-based motion planning on a 7-DoF Kinova Gen3.
 - Prototyped obstacle-avoidance groundwork in MuJoCo, training the arm to reach goals among static and moving obstacles using reward shaping and curriculum learning.
- Research Assistant**, Institute of Smart Systems and Artificial Intelligence, Nazarbayev University, Kazakhstan Jan 2023 – Jul 2024
- Built a 5.1 million-image fisheye dataset using non-linear geometric mapping and trained YOLO models on it for object recognition.
 - Generated synthetic training data for object recognition using NVIDIA Omniverse.
 - Collected hemispherical-camera images into a 1,000-image fisheye benchmark for evaluation.

Publications

Refereed Journal Papers

- [1] A. Papanikolaou, V. Sivtsov, **M. Alaran**, D. Shkolnik, E. Zereik, I. Markovic, I. Petrovic and F. Bonsignorio, "A time and object aware transformer-based method for structured hierarchical gesture recognition," *Robotics and Autonomous Systems*, Art. no. 105518, 2026, doi: 10.1016/j.robot.2026.105518 [DOI]
- [2] A. Keutayeva, C. Jesse Nwachukwu, **M. Alaran**, Z. Otarbay and B. Abibullaev, "Neurotechnology in Gaming: A Systematic Review of Visual Evoked Potential-Based Brain-Computer Interfaces," *IEEE Access*, vol. 13, pp. 74944-74966, 2025, doi: 10.1109/ACCESS.2025.3564328 [DOI]
- [3] **M. A. Alaran**, S. K. Lawal, M. H. Jiya, S. A. Egya, M. M. Ahmed, A. Abdulsalam, U. A. Haruna, M. K. Musa and D. E. Lucero-Prisno, III, "Challenges and opportunities of artificial intelligence in African health space," *Digital Health*, vol. 11, Art. no. 20552076241305915, 2025, doi: 10.1177/20552076241305915 [DOI]

Refereed Conference Papers

- [1] V. Sivtsov, **M. Alaran**, D. Shkolnik, A. Papanikolaou, I. Markovic, E. Zereik, I. Petrovic and F. Bonsignorio, "Spatially Aware Value Fusion in Decomposed Reward Architectures for Marine Manipulation," in *Proc. 23rd IFAC World Congress*, Busan, Republic of Korea, 2026.
- [2] F. Olanrewaju, M. Shafiullah, M. Al-Dhaifallah and **M. Alaran**, "APO-Optimized ARX-Based System Identification of a Ship Maneuvering System," in *2025 IEEE 22nd International Multi-Conference on Systems, Signals & Devices (SSD)*, Monastir, Tunisia, 2025, pp. 660-664, doi: 10.1109/SSD64182.2025.10989967 [DOI]
- [3] **M. Alaran**, Z. Balgabekova, and H. A. Varol, "Field of View Invariant Object Recognition Using Non-Linear Transformation Augmentation," in *IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society*, Chicago, IL, USA, 2024, pp. 1-6, doi: 10.1109/IECON55916.2024.10905287 [DOI]
- [4] U. Sharipov, S. Kasenov, **M. Alaran**, A. Askhatova, Y. Orynbay, and P. Jamwal, "Cloud-Integrated Navigation System for Scalable Autonomous Ground Robots," in *30th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, Leeds, United Kingdom, 2024, pp. 1-6, doi: 10.1109/M2VIP62491.2024.10746213 [DOI]
- [5] Z. Balgabekova, **M. Alaran**, and H. A. Varol, "A Data-Centric Approach for Object Recognition in Hemispherical Camera Images," in *IECON 2023 - 49th Annual Conference of the IEEE Industrial Electronics Society*, Singapore, 2023, pp. 1-6, doi: 10.1109/IECON51785.2023.10311869 [DOI]

Open Source

- Contributor, open-planning/roboplan** – Pinocchio-based motion planning library GitHub
- Six merged pull requests adding dual end-effector support, a Cartesian trajectory type, acceleration and jerk limits, Jacobian base-frame support, and keyboard teleoperation.
- finetune-octo-kinova-hybrid** – Octo fine-tuning pipeline for Kinova Gen3 GitHub
- Released pipeline (v1.0.0) with data collection, hand-eye calibration, training, and live-control code, plus documentation.
- wide-fov-dataset-toolkit** – Fisheye dataset generation for object detection GitHub
- Converts perspective object-detection datasets (COCO, Objects365) into fisheye and wide-FoV training data via eight lens-distortion transforms with matching bounding-box remapping; generalizes the IECON 2024 pipeline into a pip-installable, tested package.

Professional Experience

- Robotics Consultant**, IQ Academy, Ibadan, Nigeria. Jan 2022 – Jun 2022
- Built out an educational robotics lab for high school students and mentored them in designing low-cost robotics projects.

Robotics Developer (Graduate Intern) and STEM Instructor, Robotics and Artificial Intelligence Nigeria (RAIN), Ibadan, Nigeria.

Jun 2021 – May 2022

- Built an autonomous ground robot using ROS and LiDAR; delivered projects across embedded systems and IoT.
- Taught embedded C and Arduino sensor/actuator interfacing, culminating in students building a programmable robot car.

Support Engineer, Intern, Nimbles Engineering Company, Ibadan, Nigeria.

Apr 2019 – Sep 2019

- Configured and installed Siemens and Allen-Bradley PLCs to control industrial pumps, alarms, blowers, and motors; troubleshoot faulty control systems.

Additional Experience

Conference Service: Reviewer for ICRA 2027, IROS 2025, IECON 2023.

Automation Team Lead, NU Rover, Nazarbayev University (2023–2024): Led autonomous navigation, object detection, and motion control development for the pioneer NU team at the University Rover Challenge.

Oxford Machine Learning Summer School 2024: Representation learning for vision, language, and multi-modal data.

Teaching Assistant, Dept. of Robotics, Nazarbayev University (2023): Electrical and Electronic Circuits II; Microcontrollers with Lab.

President, National Society of Black Engineers, University of Ilorin Chapter (2019/2020): Led professional workshops and orientation programs for student members and incoming engineering students.

Awards and Honors

Abay Kunanbayev Scholarship

2022 – 2024

Competitive award for outstanding international students pursuing graduate degrees at Nazarbayev University.

Award of Appreciation for Excellence in Leadership and Community Building

2024

Awarded by the International Students and Scholars Services, Nazarbayev University, to students who have made significant contributions to building a vibrant and inclusive campus community.

Award of Recognition for Outstanding and Exemplary Leadership

2020

Awarded to individuals who exhibited outstanding and exemplary leadership in advancing the mission and values of NSBE at the University of Ilorin.

Award of Academic Excellence, NSBE - University of Ilorin

2018

Awarded to students who achieved exceptional academic performance and demonstrated a strong commitment to excellence in engineering studies.

Skills

Languages: Python, C/C++ , MATLAB.

Robot Learning and ML: PyTorch, JAX/Flax, Octo, LeRobot, TensorFlow Datasets (RLDS), OpenCV.

Robotics and Tools: ROS/ROS 2 (Nav2), Pinocchio, Kinova Kortex API, Linux, Git, Docker.

Hardware: Kinova Gen3 (single- and dual-arm), Unitree G1 and R1 humanoids, dexterous hands, Intel RealSense, PICO 4 XR, NVIDIA Jetson.

Simulation: MuJoCo, ManiSkill, Isaac Sim, NVIDIA Omniverse, Gazebo.